Eric Seo

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Education

Cornell University, College of Engineering, Ithaca, NY **Expected May 2028**
Bachelors of Science, Electrical and Computer Engineering **GPA: 3.69/4.00**

Relevant Experiences

CUXR (Cornell University eXtended Reality) **January 2025 – Present**
Full Team Lead, Founding Member *Ithaca, NY*

- Managing team of ~30 members creating haptic feedback systems for VR use, creating custom models for object recognition and memory in AR glasses, and signal processing and EEG work for use in BCI applications.
- Contributes hands-on engineering work, including programming MCUs with C/C++, creating various custom PCBs for motor/power systems, and interfacing with computer vision software to track user location.
- Directly managing projects such as a haptic glove system, weather simulator, and an omni-directional treadmill.
- Hosted information sessions with companies such as Valve, General Motors, and General Matter.

Picogrid Hardware Engineering **May 2026 – August 2026**
Hardware Intern *El Segundo, CA*

- Owned end-to-end mixed-signal PCB design for complex 8-layer boards, including controlled-impedance differential routing, signal-integrity analysis, EMI mitigation, and high-speed data transfer.
- Drove development of USB-C Power Delivery and STM32-based hardware, from schematic design to layout.
- Led project scoping from requirements through prototype, defining technical approaches and execution plans.

Kiewit Power Engineering **May 2025 – August 2025**
Power Engineer *Olathe, KS*

- Supported multiple projects at different stages including commissioning and plant closeout for electrical systems.
- Performed circuit design, cable sizing, and voltage drop calculations to ensure systems met electrical standards.
- Updated and maintained key documentation including one-line diagrams and schematics to keep records accurate and aligned with field conditions.

Projects

Omnidirectional Treadmill **April 2026 - Present**
Cornell XR

- Led 360° platform design, setting mechanical, structural, and user-safety requirements for VR treadmill use.
- Designing a buck-converter PCB and MCU control system to drive treadmill motion from live camera input.
- Developing a CMU OpenPose pipeline for real-time user motion tracking and closed-loop treadmill control.

DEX (Dextrous Hand) **February 2026**
Cornell Makeathon 2026 (Best Hardware Winner)

- Designed DEX, a wearable hand-tracking peripheral for real-time motion sensing and tactile interaction.
- Built the full embedded electronics stack, wiring up dual rotary encoders, a 9-DOF IMU, and a thumb switch to an ESP32 and getting motion data over I²C/SPI.
- Wrote firmware to read calibrated sensor data to a web platform as usable finger-tracking data for rehab and accessibility applications.

Skills

Relevant Coursework: Circuit Analysis, Digital Logic & Computer Organization, Microelectronics, Signals and Systems, EM Fields and Waves, Embedded Systems, Linear Algebra, Differential Equations, Foundations of Robotics.
Skills/Tools: KiCad, Altium, EMI Analysis, PCB Design, C++, C, Python, SystemVerilog, LTSpice, LaTeX, Git, STM32, ROS, Soldering, Oscilloscopes, Multimeters, Function Generators, I2C, SPI, UART, USB-C PD